

IBM HR Analytics Employee Attrition Dataset

Data description

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1. Dataset Title

IBM HR Analytics Employee Attrition & Performance Dataset.

2. Dataset Source

The dataset was obtained from Kaggle and was originally created by IBM data scientists for educational and analytical purposes.

3. Business Domain

Human Resources (HR) Analytics.

4. Dataset Size

Item	Value
Rows	1476
Columns	35
Target Variable	Attrition

Table 1: Dataset Overview

5. Description of Variables

Variable	Description
Attrition	Indicates whether the employee left the company (Yes) or stayed (No).
Age	Age of the employee in years.
BusinessTravel	Frequency of business travel (Non-Travel, Travel Rarely, or Travel Frequently).
DailyRate	Employee's daily salary rate.
Department	Department where the employee works (Sales, Research & Development, Human Resources).
DistanceFromHome	Distance between the employee's residence and workplace.
Education	Education level ranging from 1 (Below College) to 5 (Doctor).
EducationField	Employee's field of study (Medical, Life Sciences, Marketing, etc.).
EmployeeCount	Constant value representing employee count (always 1).
EmployeeNumber	Unique identification number assigned to each employee.

Variable	Description
EnvironmentSatisfaction	Employee satisfaction with the work environment (1 = Low, 4 = Very High).
Gender	Employee gender (Male or Female).
HourlyRate	Employee's hourly pay rate.
JobInvolvement	Employee involvement in work (1 = Low, 4 = Very High).
JobLevel	Job hierarchy level ranging from 1 to 5.
JobRole	Employee's specific job title.
JobSatisfaction	Overall satisfaction with the job (1 = Low, 4 = Very High).
MaritalStatus	Employee marital status (Single, Married, Divorced).
MonthlyIncome	Monthly salary of the employee.
MonthlyRate	Monthly billing rate.
NumCompaniesWorked	Number of companies worked for before joining IBM.
Over18	Indicates whether the employee is over 18 years old.
OverTime	Indicates whether the employee works overtime.
PercentSalaryHike	Percentage increase in salary compared with the previous year.
PerformanceRating	Employee performance evaluation rating.
RelationshipSatisfaction	Satisfaction with workplace relationships (1 = Low, 4 = Very High).
StandardHours	Standard working hours (constant value = 80).
StockOptionLevel	Level of stock options provided to the employee.
TotalWorkingYears	Total years of professional work experience.
TrainingTimesLastYear	Number of training sessions attended during the previous year.
WorkLifeBalance	Employee rating of work-life balance (1 = Bad, 4 = Excellent).
YearsAtCompany	Number of years worked at the current company.
YearsInCurrentRole	Years spent in the current job role.
YearsSinceLastPromotion	Years since the employee received the last promotion.
YearsWithCurrManager	Years working with the current manager.

6. Reason for Selecting the Dataset

This dataset was selected because employee retention is one of the most critical challenges faced by organizations. The dataset combines demographic, financial, job-related, and satisfaction variables, enabling comprehensive exploratory data analysis and meaningful business insights.

7. Business Problem

Employee attrition is costly for organizations because it increases recruitment expenses, training costs, and knowledge loss.

The primary objective of this project is to identify the factors that influence employee attrition and provide recommendations to improve employee retention.

8. Why This Dataset Meets the Project Requirements

This dataset satisfies the project requirements because:

- It is a structured real-world HR dataset.
- It contains 1,476 employee records and 35 variables.
- It includes both numerical and categorical variables.
- It supports KPI creation such as Attrition Rate and Average Monthly Income.
- It contains data quality issues suitable for demonstrating data cleaning techniques in Microsoft Excel.
- It enables meaningful visualization and dashboard creation.

9. Expected Business Insights

Possible insights include:

- Departments with the highest attrition.
- Job roles most likely to leave.
- Impact of overtime on employee turnover.
- Relationship between salary and attrition.
- Effect of work-life balance on employee retention.
- Influence of promotion history on employee satisfaction.

10. Conclusion

This dataset provides an excellent opportunity to perform HR analytics using Microsoft Excel. Through data cleaning, KPI creation, visualization, and dashboard development, valuable insights can be generated to support strategic HR decision-making and reduce employee turnover.

— End of Report —